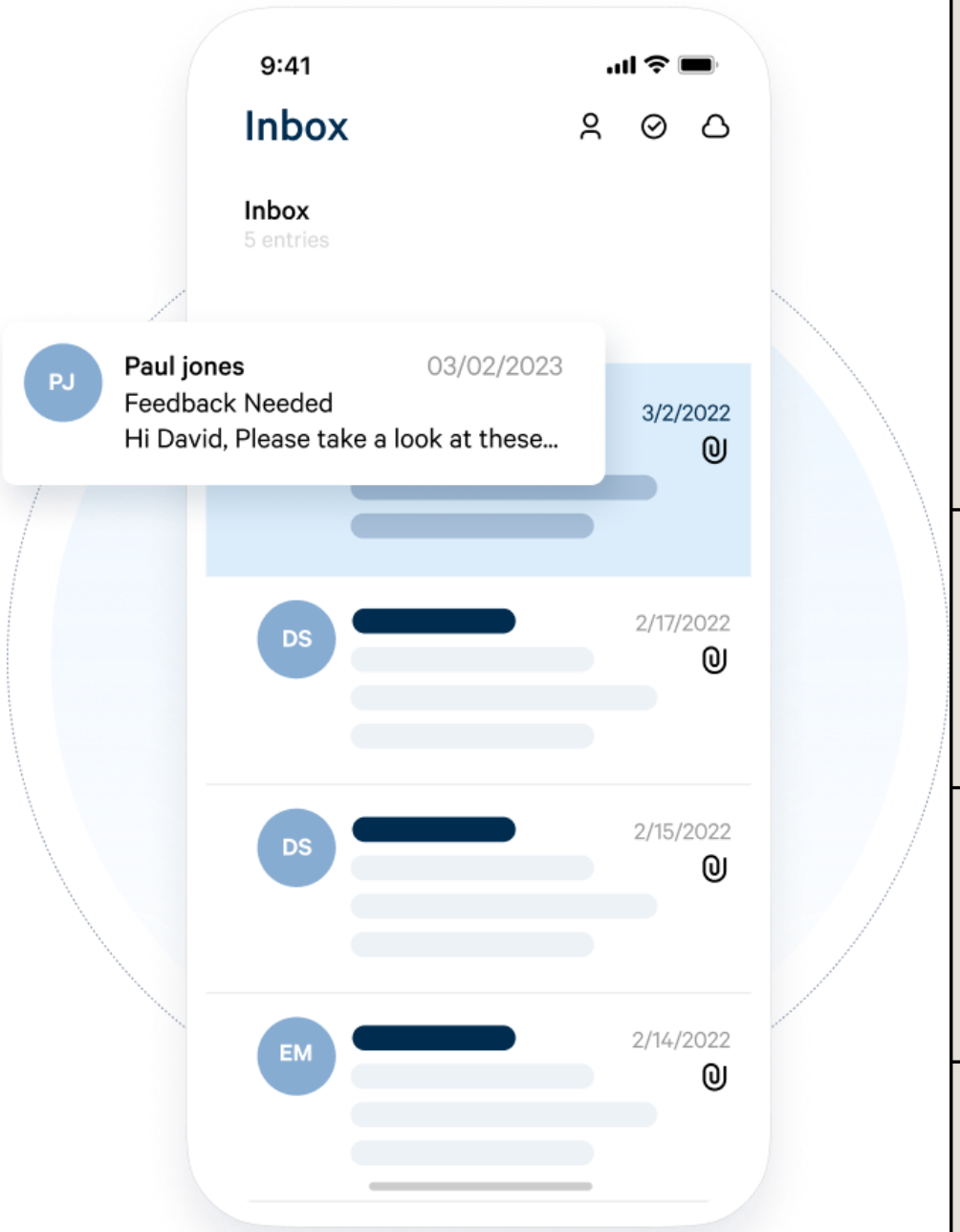
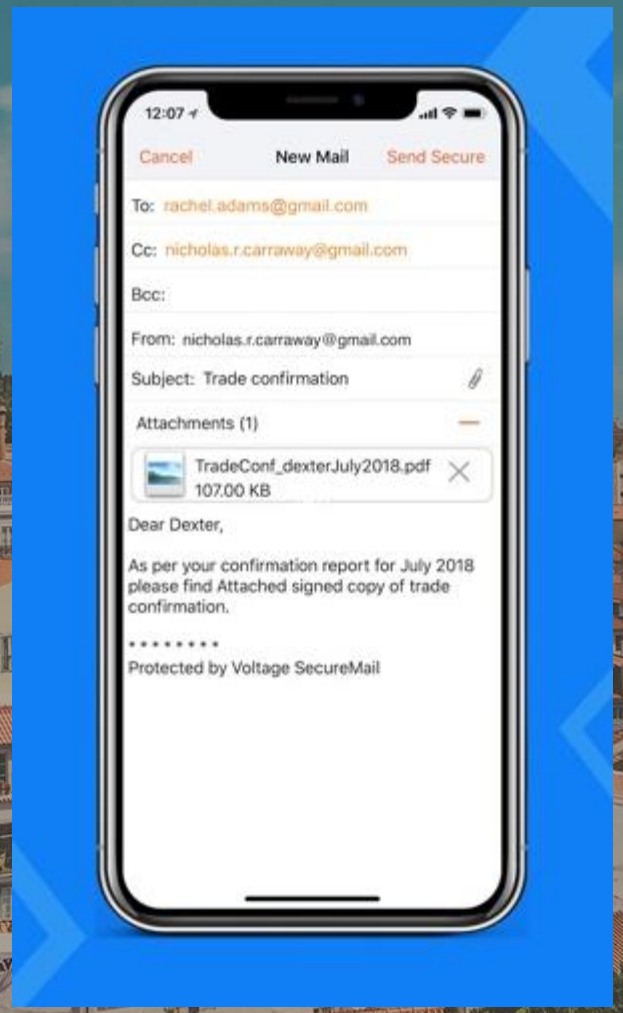


EMAIL, SECURE MAIL, ITS THREATS, DETECTION

A PRESENTATION BY NIKOLA ŠAŠINKOVÁ

PRINCIPLES OF INFORMATION SECURITY

2ND PROGRESS REPORT



AGENDA

- SPAM FILTERS
- FORTINET ANTISPAM
- ARCHITECTURE OF THE EXPERIMENT INFRASTRUCTURE
- DNS TESTING WITH MXTOOLBOX
- TOOLS USED FOR THIS EXPERIMENT
- EXPERIMENT PERFORMED ON THE FOLLOWING SETUPS
- RESULTS OF THE EXPERIMENT
- EVALUATION OF THE RESULTS OF THE EXPERIMENT
- SOURCE CODE FOR SENDING AN EMAIL IN PYTHON
- TIPS HOW TO PREVENT SPAM MAILS AND IMPROVE SECURITY
- GENERAL TIPS ON HOW TO STAY SAFE ONLINE
- PLAN FOR COMPLETING THE PROJECT
- SOURCES

SPAM FILTERS

- Spam filters are tools that use several techniques to stop unwanted or dangerous emails before they reach the inbox
- Content filters look at the words, phrases and structure of the email
- They search for typical spam patterns, like special offers or strange formatting
- Header filters check technical details, such as the sender address and server path
- These filters can spot fake sender identities or unusual sending behavior
- Blacklist filters block emails from addresses, domains or IPs that are known sources of spam
- Rule-based filters use custom rules, for example blocking emails with risky attachments or specific keywords

SPAM FILTERS

- Advanced spam filters use machine learning or Bayesian methods to learn from past examples
- The system often gives each email a spam score based on all these checks
- If the score is too high, the email is moved to the spam folder or blocked completely
- Because spammers change tactics and use new domains, spam filters must be updated often and use current threat intelligence
- By combining different methods, spam filters reduce phishing and malware and keep user inboxes cleaner and safer

FORTINET ANTISPAM

- Security service in FortiMail and other Fortinet products to block unwanted or dangerous emails
- Uses IP reputation, URL filtering, content scanning and machine learning to detect spam and phishing
- Gets constant threat intelligence from the FortiGuard network and can block, quarantine or tag emails

IP Reputation

- FortiMail sends the sender's IP address to FortiGuard Antispam for a reputation check
- FortiGuard returns a reputation level based on known bad or spammy IP addresses
- Very bad IPs can be rejected, others can be quarantined or tagged to reduce repeat spam

URL Filtering

- FortiMail scans the email body and finds all URLs
- Each URL is checked against FortiGuard URL blacklist and blocked categories (malware, hacking, spam, …)
- If a URL is in a blocked category, the email is treated as spam and an action is applied (quarantine, reject, tag)

FORTINET ANTISPAM

Spam Outbreak Protection

- When many suspicious emails arrive from unknown sources, FortiMail can hold them in a temporary queue
- During this time, FortiGuard updates its lists and re-checks the emails with new information
- After the outbreak period, emails are re-evaluated and then rejected, quarantined or tagged

Heuristic and Rule-Based Scanning

- FortiMail uses heuristic rules that look for suspicious patterns in content and headers
- Each rule gives a score, all scores are added and compared to a spam threshold
- Admins can tune how many rules are used to balance detection quality and system performance

Bayesian Filtering

- Bayesian filter learns from past emails marked as spam or not spam
- FortiMail can use a global Bayesian database and per-domain databases
- Users can train the filter by sending example emails, so detection becomes more accurate over time

FORTINET ANTISPAM

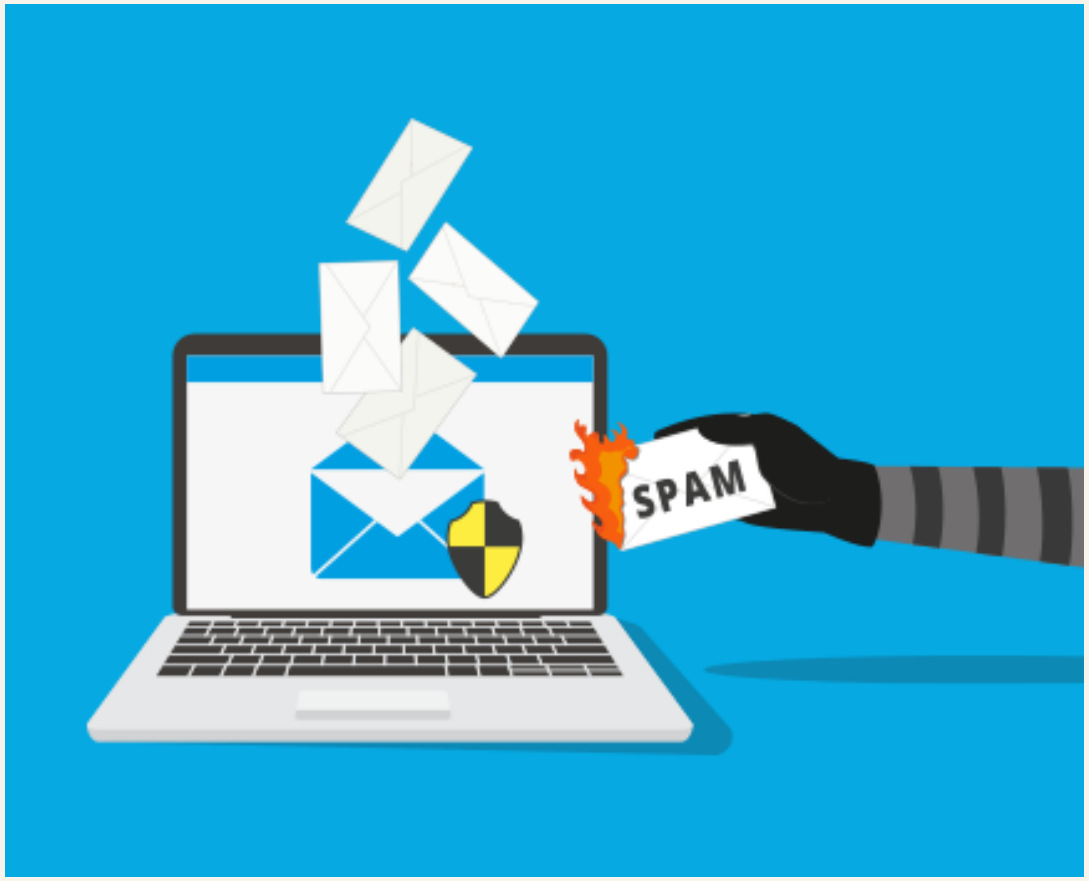


Image and PDF Spam Scanning

- FortiMail can scan embedded images (GIF, JPG, PNG) and image attachments for spam content
- In aggressive mode, it also checks attached images and the first page of PDF files
- This helps detect “image-only” or PDF spam that tries to bypass normal text-based filters

Quarantine, Tag, Reject

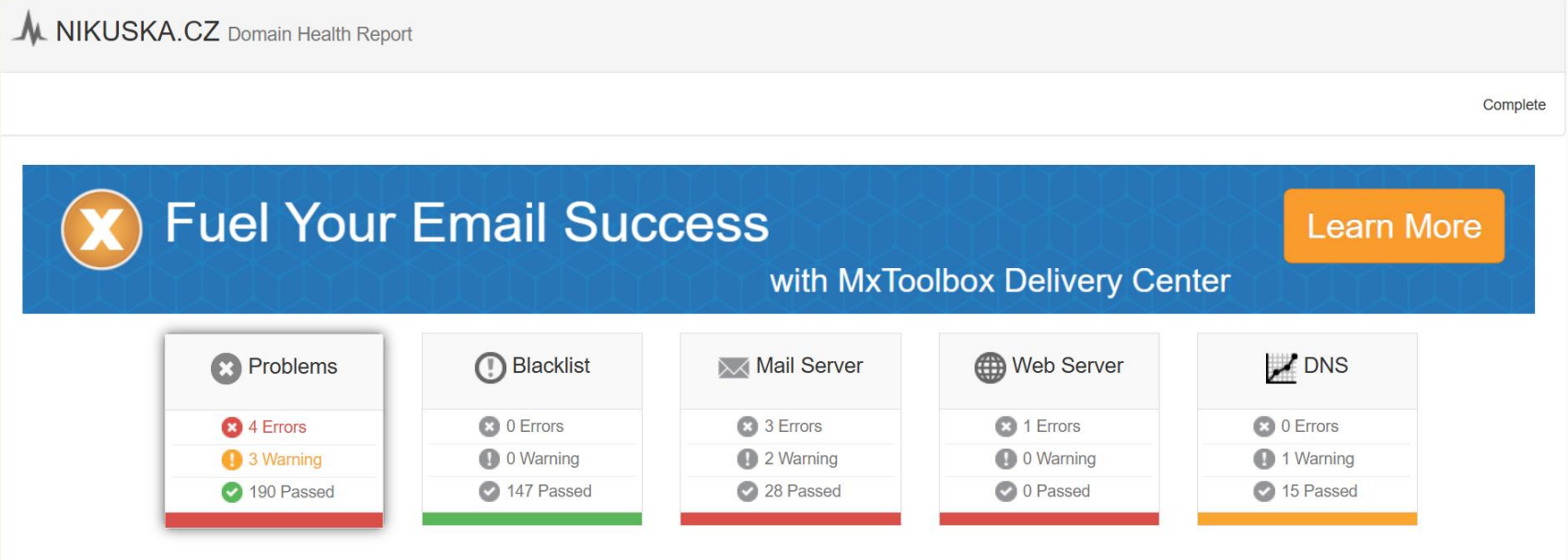
- For detected spam, FortiMail can reject, quarantine or tag the email based on action profiles
- Less certain spam can be quarantined for user review, very risky spam can be rejected at once
- Different scan types (FortiGuard, heuristic, Bayesian) can use different actions for fine-grained control

ARCHITECTURE OF THE EXPERIMENT INFRASTRUCTURE

- The experimental infrastructure is based on my own domain **nikuska.cz**
- The DNS zone for **nikuska.cz** is managed in the **emailprofi** DNS system provided by **Seznam**
- The domain is configured with the main email authentication mechanisms: **SPF**, **DKIM** and **DMARC**
- The **SPF record** defines which mail servers are allowed to send email on behalf of *nikuska.cz*
- **DKIM** is used to sign outgoing messages with a cryptographic signature that can be verified using a public key in DNS
- **DMARC** combines SPF and DKIM results and defines what the receiving server should do with failing messages *(for example none, quarantine or reject)*
- For the purposes of the experiment, I also created intentionally **incorrect SPF and DMARC configurations** to simulate common misconfigurations

DNS TESTING WITH MXTOOLBOX

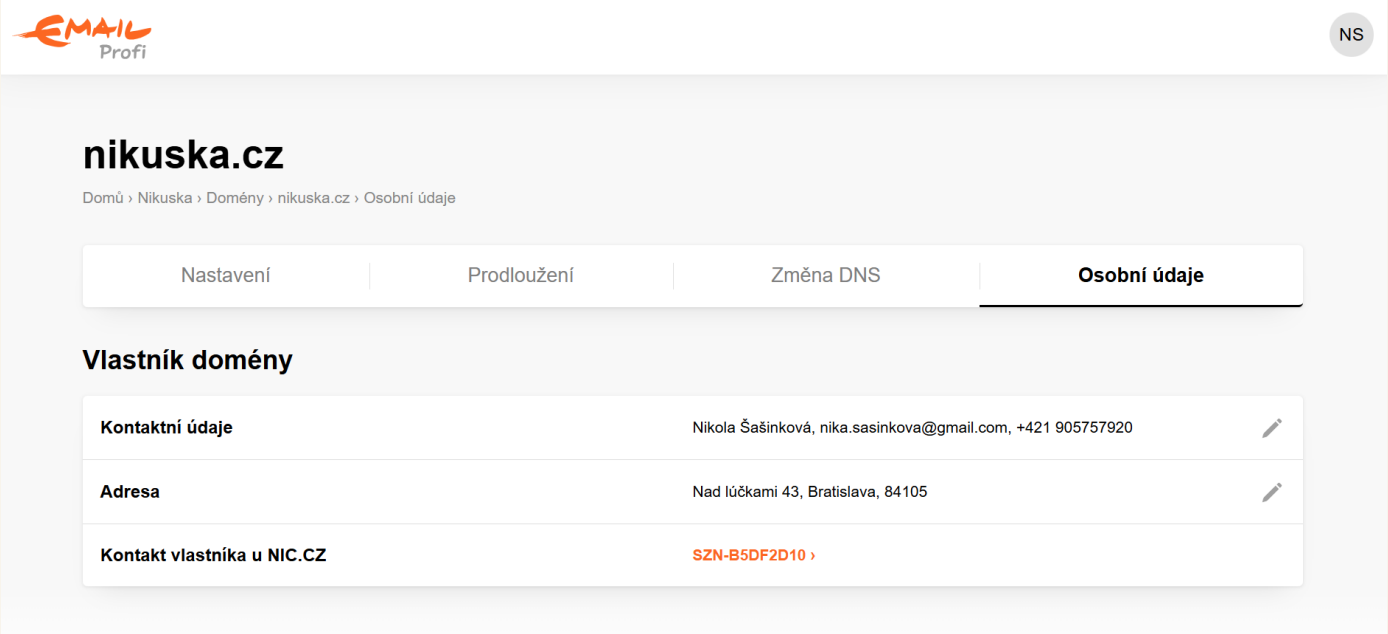
- The configuration of SPF, DKIM and DMARC is tested using **MXToolbox**
- I used the **"Email Health"** function in MXToolbox and entered the domain name **nikuska.cz**
- MXToolbox analyses DNS records related to email delivery, security and reports possible errors and warnings
- The same tests are run for both **correct** and **intentionally incorrect** DNS configurations
- The results of these tests are recorded, compared and analyzed to verify the correctness of the email infrastructure settings



TOOLS USED FOR THIS EXPERIMENT

EMAILPROFI

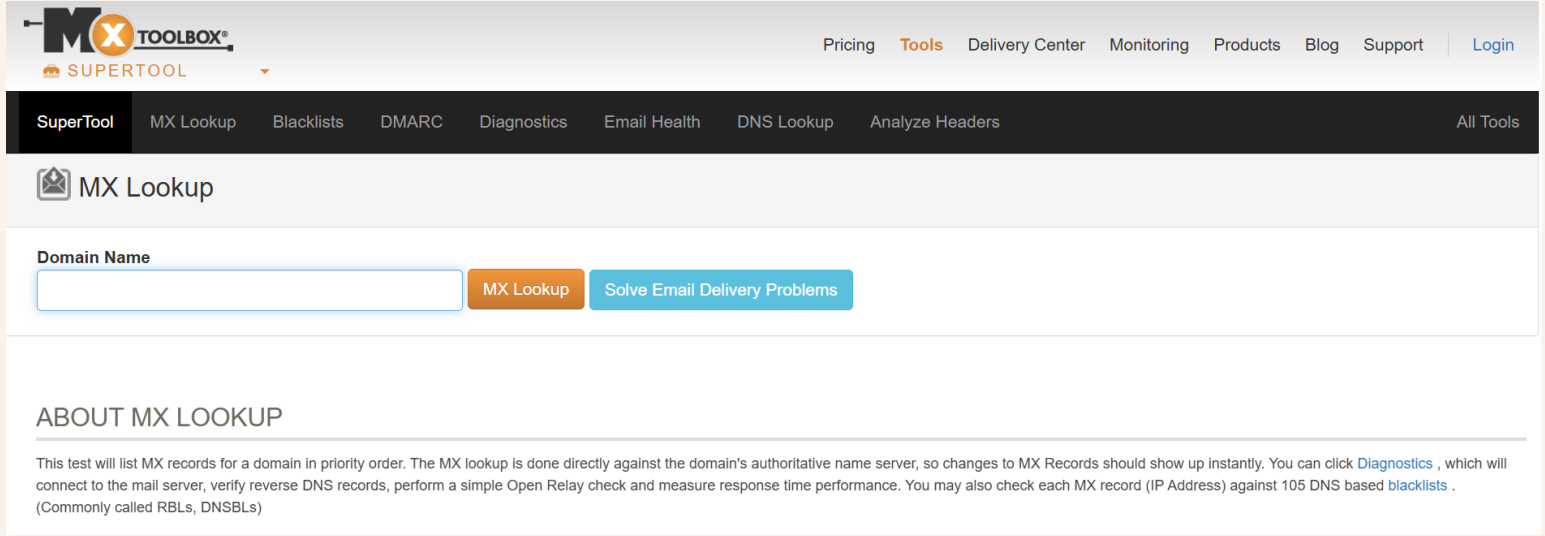
- **What it is:** Email hosting for own domain with Seznam webmail, **unlimited mailboxes & storage**, calendar, and anti-spam
- **Connect existing domain:** Sign in -> Add new domain -> enter domain -> accept license & I own a domain -> generate MX records -> update DNS -> create mailbox
- **Register new domain:** Sign in -> Add new domain -> enter new **.cz** name -> accept license -> buyer info -> purchase
- **Pros:** Unlimited users & storage, familiar Seznam interface, calendar, support for mobile & desktop clients
- **Cons:** Need to **buy a domain** if you don't have one, tied to Seznam infrastructure (if Seznam has downtime or policy changes, email could be affected), must manage **MX records and DNS correctly**, wrong **SPF/DMARC** -> mail may be flagged as spam or rejected



TOOLS USED FOR THIS EXPERIMENT

MXTOOLBOX

- **What it is:** Online toolkit for **email structure** and **domain health check** - used to check DNS configs, deliverability, blacklist status, and mail server availability (useful for both **defenders** and **attackers**)
- **Core checks: DNS record lookup** (MX,A,TXT), **SMTP diagnostics** (incl. open-relay test), **Blacklist check** (IP,domain), **Email header analysis** (human-readable format), **Monitoring & alerts** (e.g., setup alerts when domain gets blacklisted)
- **Auth focus:** Quick verification of **SPF**, **DKIM**, **DMARC** records, confirm whether third-party senders use **clean (non-blacklisted) Ips**
- **Pros:** Wide range of DNS/email tools, **real-time SMTP diagnosis**
- **Cons:** Can feel **complex** and hard to navigate
- **Plans: Free (1 domain)**, **Delivery Center ($129/mo)**, **Delivery Center Plus ($399/mo)** with advanced monitoring, threat tools, and higher message & domain limits



EXPERIMENT PERFORMED ON THE FOLLOWING SETUPS

CORRECT DNS SETUP

@	TXT	v=spf1 include:spf.seznam.cz ~all	1800
_dmarc	TXT	v=DMARC1; p=reject; pct=100; adkim=r; aspf=r	1800
szn20221014._domainkey	CNAME	szn20221014._domainkey.seznam.cz.	1800

INCORRECT SPF, CORRECT DMARC

@	TXT	v=spf1 ipv4:203.0.113.1 -all	1800
_dmarc	TXT	v=DMARC1; p=reject; pct=100; adkim=r; aspf=r	1800
szn20221014._domainkey	CNAME	szn20221014._domainkey.seznam.cz.	1800

CORRECT SPF, INCORRECT DMARC

@	TXT	v=spf1 include:spf.seznam.cz ~all	1800
_dmarc	TXT	v=DMARC1; p=rejct; pct=100; adkim=r; aspf=r	1800
szn20221014._domainkey	CNAME	szn20221014._domainkey.seznam.cz.	1800

RESULTS OF THE EXPERIMENT

INCORRECT SPF, CORRECT DMARC

- Warning: **Reverse DNS does not match the SMTP banner** (hostname in banner **is not** PTR hostname) - the SMTP banner should show the **real server hostname** that matches reverse DNS, to increase trust
- MXToolBox **successfully connected** to the mail server for *nikuska.cz* on port 25
- The server supports standard SMTP features: **8BITMIME**, **SIZE**, **PIPELINING**, **STARTTLS**
- The server correctly **rejects non-existent mailboxes** with a 550 error
- Overall SMTP transcript shows a **healthy and correctly behaving mail server**

Mail Server

mx:nikuska.cz

Pref	Hostname	IP Address	TTL
10	1af6af85a3151459.mx2.emailprofi.seznam.cz	77.75.77.146	30 min
10	1af6af85a3151459.mx2.emailprofi.seznam.cz	2a02:598:2::1146	30 min
20	1af6af85a3151459.mx1.emailprofi.seznam.cz	77.75.77.56	30 min
20	1af6af85a3151459.mx1.emailprofi.seznam.cz	2a02:598:2::1056	30 min

dmarc:nikuska.cz

	Category	Host	Result	
✓	dmarc	nikuska.cz	DMARC Record found	More Info
✓	dmarc	nikuska.cz	The record is valid	More Info
✓	dmarc	nikuska.cz	Multiple DMARC records corrected to a single record.	More Info
✓	dmarc	nikuska.cz	DMARC Quarantine/Reject policy enabled	More Info
✓	dmarc	nikuska.cz	All external domains in your DMARC record are giving permission to send them DMARC reports.	More Info

smtp:1af6af85a3151459.mx2.emailprofi.seznam.cz

	Category	Host	Result	
!	smtp	1af6af85a3151459.mx2.emailprofi.seznam.cz	Reverse DNS does not match SMTP Banner	More Info
✓	smtp	1af6af85a3151459.mx2.emailprofi.seznam.cz	OK - 77.75.77.146 resolves to wildcard.mx2.emailprofi.seznam.cz	More Info
✓	smtp	1af6af85a3151459.mx2.emailprofi.seznam.cz	OK - Reverse DNS is a valid Hostname	More Info
✓	smtp	1af6af85a3151459.mx2.emailprofi.seznam.cz	OK - Supports TLS.	More Info
✓	smtp	1af6af85a3151459.mx2.emailprofi.seznam.cz	1.389 seconds - Good on Connection time	More Info
✓	smtp	1af6af85a3151459.mx2.emailprofi.seznam.cz	OK - Not an open relay.	More Info
✓	smtp	1af6af85a3151459.mx2.emailprofi.seznam.cz	2.249 seconds - Good on Transaction Time	More Info

RESULTS OF THE EXPERIMENT

INCORRECT SPF, CORRECT DMARC

- SPF check found a **syntax error** in the SPF record
- Incorrect SPF syntax can **weaken email authentication** and cause delivery or spam-filtering issues
- DNS health report: *nikuska.cz* has a valid DNS record
- The NS list matches the **parent .cz zone**, and SOA timing values are **within recommended ranges**
- DNS trace shows correct **delegation from .cz name servers to emailprofi.cz**, confirming proper DNS setup

spf:nikuska.cz - 1 Tests Failed

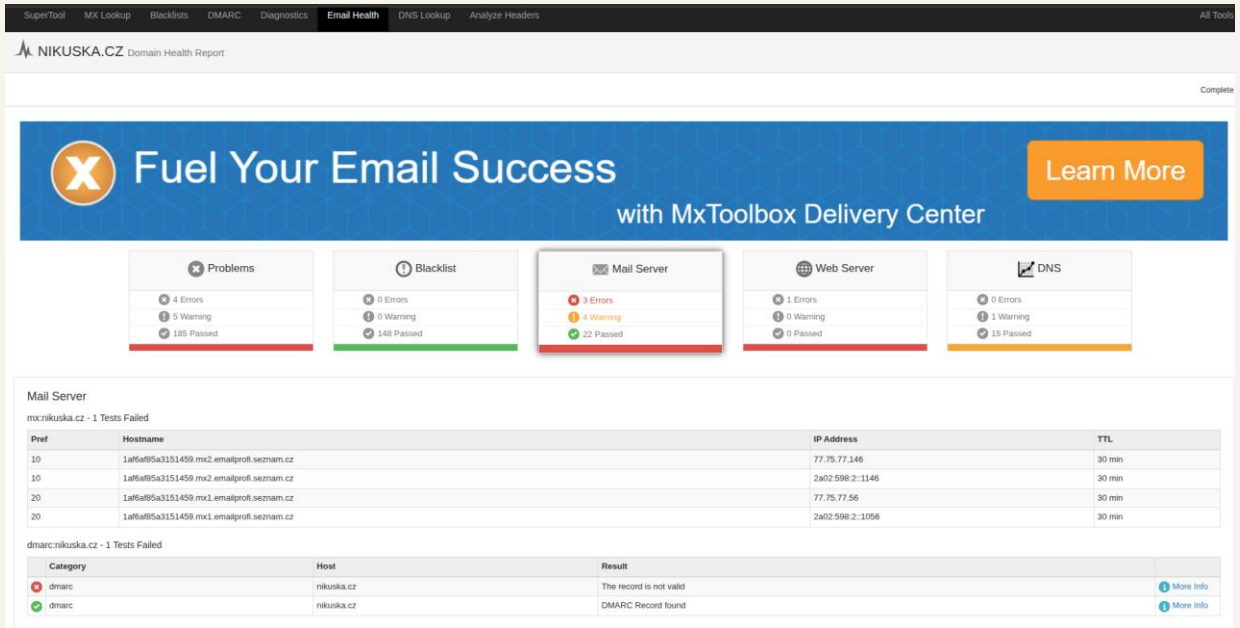
	Category	Host	Result	
✗	spf	nikuska.cz	Invalid syntax found	More Info
✓	spf	nikuska.cz	SPF Record found	More Info
✓	spf	nikuska.cz	No deprecated records found	More Info
✓	spf	nikuska.cz	Less than two records found	More Info
✓	spf	nikuska.cz	No items after 'ALL'.	More Info
✓	spf	nikuska.cz	DMARC Record found	More Info
✓	spf	nikuska.cz	DMARC Quarantine/Reject policy enabled	More Info

Type	Domain Name	TTL	Record	Prefix	Value	PrefixDesc	Description
record					txt		v=spf1 ipv4:203.0.113.1 -all
v					spf1		The SPF record version
ipv4				+	203.0.113.1	Pass	Unknown
all				-		Fail	Always matches. It goes at the end of your record.

RESULTS OF THE EXPERIMENT

CORRECT SPF, INCORRECT DMARC

- **DMARC record** exists, but its **syntax is invalid**, so the DMARC test fails
- The first SMTP test reports **two warnings**:
  - Reverse DNS does not match the SMTP banner
  - Server **may be an open relay**
- Other SMTP checks are correct: Reverse DNS works, **TLS is supported**, and IPv4 resolves properly
- The SMTP session transcript shows support for **8BITMIME**, **PIPELINING** and **STARTTLS**
- The server accepted the **envelope sender** used by the test tool
- The connection from the testing IP was dropped, which caused the **"may be an open relay"** warning



RESULTS OF THE EXPERIMENT

CORRECT SPF, INCORRECT DMARC

- The **SPF record** for nikuska.cz is present and its syntax is valid
- No deprecated mechanisms, no loops and no excessive DNS lookups were found in SPF
- SPF correctly **authorizes Seznam servers** to send email for nikuska.cz
- DNS health is good overall, the only warning is **"SOA Serial Number Format is Invalid"** - the SOA serial value works functionally but does not use the recommended **YYYYMMDDnn** style

Dns Server

dns:nikuska.cz

	Category	Host	Result	
!	dns	nikuska.cz	SOA Serial Number Format is Invalid	More Info
✓	dns	nikuska.cz	DNS Record found	More Info
✓	dns	nikuska.cz	No Bad Glue Detected	More Info
✓	dns	nikuska.cz	At Least Two Name Servers Found	More Info
✓	dns	nikuska.cz	All name servers are responding	More Info
✓	dns	nikuska.cz	All of the name servers are Authoritative	More Info
✓	dns	nikuska.cz	Local NS list matches Parent NS list	More Info
✓	dns	nikuska.cz	Name Servers appear to be Dispersed	More Info
✓	dns	nikuska.cz	Name Servers have Public IP Addresses	More Info
✓	dns	nikuska.cz	Serial numbers match	More Info
✓	dns	nikuska.cz	Primary Name Server Listed At Parent	More Info
✓	dns	nikuska.cz	SOA Refresh Value is within the recommended range	More Info
✓	dns	nikuska.cz	SOA Retry Value is within the recommended range	More Info
✓	dns	nikuska.cz	SOA Expire Value within recommended limits	More Info
✓	dns	nikuska.cz	SOA Minimum TTL Value is within allowed values	More Info
✓	dns	nikuska.cz	No Open Recursive Name Server Detected	More Info

DNS

Type	Domain Name	IP Address	TTL	Time (ms)	Status	Auth	Parent	Local
NS	ams.emailprofi.cz	77.75.75.166	30 min	116	✓	✓	✓	✓
NS	ans.emailprofi.cz	77.75.74.238	30 min	111	✓	✓	✓	✓

RESULTS OF THE EXPERIMENT
CORRECT SPF, CORRECT DMARC

- Mail Server section now shows **no errors** - configuration is correct
- All email for **nikuska.cz** is handled by **Seznam's mail servers**, which support both IPv4 and IPv6
- **DMARC** is present for the domain
- DMARC syntax is **correct**, and a **quarantine policy** is enabled
- SMTP check on the MX server shows **one warning** - reverse DNS does not match the SMTP banner
- Other SMTP parameters are properly configured: Reverse DNS is a **valid hostname**, **TLS is supported**, and connection/transaction times are good
- SMTP diagnostic transcript: MXToolBox **successfully connected** to the mail server
- The server supports standard SMTP features: **8BITMIME, SIZE, PIPELINING, STARTTLS**
- The server correctly **rejects non-existent mailboxes** with a 550 code
- DNS lookup time is around **3.2 seconds**, which is acceptable
- These results confirm **properly functioning mail receiving** and a **healthy SMTP server**
- SPF checks show **no errors** and **100% correct configuration**
- The full DNS resolution process completes in about **1975 ms**, confirming a stable DNS setup

RESULTS OF THE EXPERIMENT

CORRECT SPF, CORRECT DMARC

SuperTool | MX Lookup | Blacklists | DMARC | Diagnostics | Email Health | DNS Lookup | Analyze Headers | All Tools

NIKUSKA.CZ Domain Health Report

Complete

Get the Best in Email Delivery
with MxToolbox Delivery Center
Learn More

Problems	Blacklist	Mail Server	Web Server	DNS
1 Errors	0 Errors	0 Errors	1 Errors	0 Errors
3 Warning	0 Warning	2 Warning	0 Warning	1 Warning
197 Passed	148 Passed	34 Passed	0 Passed	15 Passed

4 Problems

Category	Host	Result	
http	nikuska.cz	The remote name could not be resolved: 'nikuska.cz' (http://nikuska.cz)	More Info
dns	nikuska.cz	SOA Serial Number Format is Invalid	More Info
smtp	1af6af85a3151459.mx2.emailprofi.seznam.cz	Reverse DNS does not match SMTP Banner	More Info
smtp	1af6af85a3151459.mx1.emailprofi.seznam.cz	Reverse DNS does not match SMTP Banner	More Info

Blacklists

blacklist:nikuska.cz

Category	Host	Result	
blacklist	nikuska.cz	ivmURI	More Info
blacklist	nikuska.cz	Nordspam DBL	More Info

RESULTS OF THE EXPERIMENT

CORRECT SPF, CORRECT DMARC

Session Transcript:

```
TimeStamp:

- LookupServer [dns:nikuska.cz] 1975ms

Depth: 1
ServerName: d.ns.nic.cz
ServerIP: 193.29.206.1
Authoritative: NON-AUTH
ElapsedTime: 94 ms
Result: Received 2 Referrals , rcode=NO_ERROR
Question:
Answers: nikuska.cz. 3600 IN NS ams.emailprofi.cz,nikuska.cz. 3600 IN NS ans.emailprofi.cz,

TimeStamp:
Depth: 2
ServerName: ans.emailprofi.cz
ServerIP: 77.75.74.238
Authoritative: AUTH
ElapsedTime: 122 ms
Result: Received 2 Answers , rcode=NO_ERROR
Question:
Answers: nikuska.cz. 1800 IN NS ans.emailprofi.cz,nikuska.cz. 1800 IN NS ams.emailprofi.cz,
```

Session Transcript:

```
Transcript: Connecting to 77.75.77.56

Transcript:

Transcript: 220 email.seznam.cz ESMTP [1307 ms]

Transcript: EHLO keeper-us-east-1d.mxtoolbox.com

Transcript: 250-email.seznam.cz email zdarma na cely zivot (session-id:b7d53a00-545c-4902-9573-93e114e0c092)

Transcript: 250-8BITMIME

Transcript: 250-PIPELINING

Transcript: 250-SIZE 52428800

Transcript: 250-STARTTLS

Transcript: 250 ENHANCEDSTATUSCODES [247 ms]

Transcript: MAIL FROM:<supertool@mxtoolboxsmtpdiag.com>

Transcript: 250 2.1.0 ok [327 ms]

Transcript: RCPT TO:<test@mxtoolboxsmtpdiag.com>

Transcript: 550 5.1.1 sorry, no such mailbox here [259 ms]

Transcript:

Transcript:

Transcript: - LookupServer [smtp:1af6af85a3151459.mx1.emailprofi.seznam.cz] 3431ms

Transcript:
```

EVALUATION OF THE RESULTS OF THE EXPERIMENT

COMPARISON OF MAIL SERVER HEALTH RESULTS:

	ERRORS	WARNINGS	PASSED
CORRECT SPF, CORRECT DMARC	0	2	34
INCORRECT SPF, CORRECT DMARC	1	2	26
CORRECT SPF, INCORRECT DMARC	3	4	22

EVALUATION OF THE RESULTS OF THE EXPERIMENT

COMPARISON OF MAIL SERVER HEALTH RESULTS:

- I compared the three setups by the **number of errors, warnings and passed checks** in the MXToolBox reports
- With the **correct setup**, there were only **two warnings**, both saying that **reverse DNS does not contain the SMTP banner**, while all other checks passed
- With **incorrect SPF**, there was **one error** (SPF - invalid syntax found) and **two warnings** (both about reverse DNS not matching the SMTP banner)
- With **incorrect DMARC**, there were three **errors:** SPF - recommended to use quarantine or reject policy, the **DMARC record is invalid**, and there was also an **"http could not be resolved"** error (not important for this experiment)
- In the incorrect DMARC case there were **four warnings in total**: Two **"reverse DNS does not match SMTP Banner"** warnings, and two **"may be an open relay"** warnings

EVALUATION OF THE RESULTS OF THE EXPERIMENT

COMPARISON OF DNS HEALTH RESULTS:

	ERRORS	WARNINGS	PASSED
CORRECT SPF, CORRECT DMARC	0	1	15
INCORRECT SPF, CORRECT DMARC	0	1	15
CORRECT SPF, INCORRECT DMARC	0	1	15

- During this experiment, the setup did not affect the DNS health results and there was exactly one warning, **"SOA Serial Number Format is invalid"**, which was the same for all setups

SOURCE CODE FOR SENDING AN EMAIL IN PYTHON

- I used **Python 3.12**, **PyCharm** and Python libraries (email and smtplib)
- I authenticated to **SMPT gmail.com** using an **App Password**, which is a 16-character password generated for this purpose

- The Python program consists of **two functions**
  - The first function creates an email from arguments, sets **SENDER**, **SUBJECT**, **RECIPIENTS** and adds attachments one by one as bytes
  - The second function sends the already created email: It connects to the SMTP server, sets up encryption, authenticates me and then sends the message
- In the main part of the program, it is only necessary to set **SENDER**, **SUBJECT**, **RECIPIENTS**, **BODY_TEXT**, **ATTACHMENTS** and call these two functions to successfully send an email

(Result) Using this code, I am able to send an email with attachments
(Evaluation) It is advantageous because all parameters of the email are defined directly in the code, so it is very easy to send the exact same email multiple times, and if someone wants to change only one detail (for example the subject or one attachment), he or she just modifies a single value in the code instead of manually creating and clicking through a new email each time

SOURCE CODE FOR SENDING AN EMAIL IN PYTHON

```python
import ssl
import mimetypes
import smtplib
from pathlib import Path
from email.message import EmailMessage
import getpass

SMTP_HOST = "smtp.gmail.com"
SMTP_PORT = "587"
SMTP_USER = "nika.sasinkova@gmail.com"
SMTP_PASSWORD = getpass.getpass("Zadaj heslo od Gmailu alebo App Password ")

if not SMTP_USER or not SMTP_PASSWORD:
    exit()

def create_email(subject, sender, recipients, body_text, attachment_paths: list):
    msg = EmailMessage()  # prazdny email
    msg["Subject"] = subject
    msg["From"] = sender
    msg["To"] = ", ".join(recipients)

    msg.set_content(body_text)

    if attachment_paths is not None:
        for p in attachment_paths:

            if not p.is_file():
                raise FileNotFoundError(f"Priloha nebola najdena: {p}")

            mime_type, _ = mimetypes.guess_type(p)
            if mime_type is None:
                mime_type = "application/octet-stream"

            maintype, subtype = mime_type.split("/", 1)
            data = None
            with p.open("rb") as fp:
                data = fp.read()

            msg.add_attachment(
                data,
                maintype=maintype,
                subtype=subtype,
                filename=p.name,
            )

    return msg

```

```python
def send_email(message):
    print("Posielam email")
    context = ssl.create_default_context()
    print("Pripajam sa do %s:%s", SMTP_HOST, SMTP_PORT)

    with smtplib.SMTP(SMTP_HOST, SMTP_PORT) as server:
        server.starttls(context=context)
        server.login(SMTP_USER, SMTP_PASSWORD)
        server.send_message(message)

    print(f"Sprava poslana: {message['To']}")


if __name__ == "__main__":
    SUBJECT = "Test"
    SENDER = "nika.sasinkova@gmail.com"
    RECIPIENTS = ["nikolka.sasinkova@gmail.com"]
    BODY_TEXT = """\
Test posielania emailu.
"""

    ATTACHMENTS = [
        Path("C:/Users/nikola/Pictures/Screenshots/test.png")
    ]

    msg = create_email(
        subject=SUBJECT,
        sender=SENDER,
        recipients=RECIPIENTS,
        body_text=BODY_TEXT,
        attachment_paths=ATTACHMENTS,
    )

    send_email(msg)
```

TIPS HOW TO PREVENT SPAM MAILS AND IMPROVE SECURITY

- **Avoid unknown links and attachments:** Do not blindly click any links or open attachments, confirm the message by contacting the sender through a known and trusted methods
- **Flag spam messages:** Use “mark as spam” or “report” feature, this helps improve spam filters for you and other users
- **Keep software up to date:** Regularly update your email client, antivirus tools and operating system
- **Use Two-Factor authentication:** Enable 2FA on your accounts to add an extra security layer

GENERAL TIPS ON HOW TO STAY SAFE ONLINE

- **Install up-to-date antivirus and firewall software:** Have reputable security software and download updates regularly
- **Create a strong password:** Mix of letters, symbols and numbers
- **Change password regularly:** It is sensible to update your password from time to time
- **Do not share passwords**
- **Be alert to online scams**: Be cautious of offers or deals that seem unusually generous or "too good to be true"

PLAN FOR COMPLETING THE PROJECT

FINAL PROJECT SUBMISSION

Comprehensive project description, explanation of the vocabulary required to understand how email works

FINAL PROJECT SUBMISSION

My setup of the required tools

FINAL PROJECT SUBMISSION

Description of the security of TLS and E2EE

FINAL PROJECT SUBMISSION

Comprehensive reports from the performed tests

FINAL PROJECT SUBMISSION

Accompanying Python source code

YEAR 2025

SOURCES

- https://is1-ssl.mzstatic.com/image/thumb/PurpleSource116/v4/e7/d1/a1/e7d1a134-6834-b54c-cc5c-24e91c706da7/2513b58d-1220-4d91-8214-1d3cdbdfd131_3.png/300x0w.jpg, 15.11.2025.
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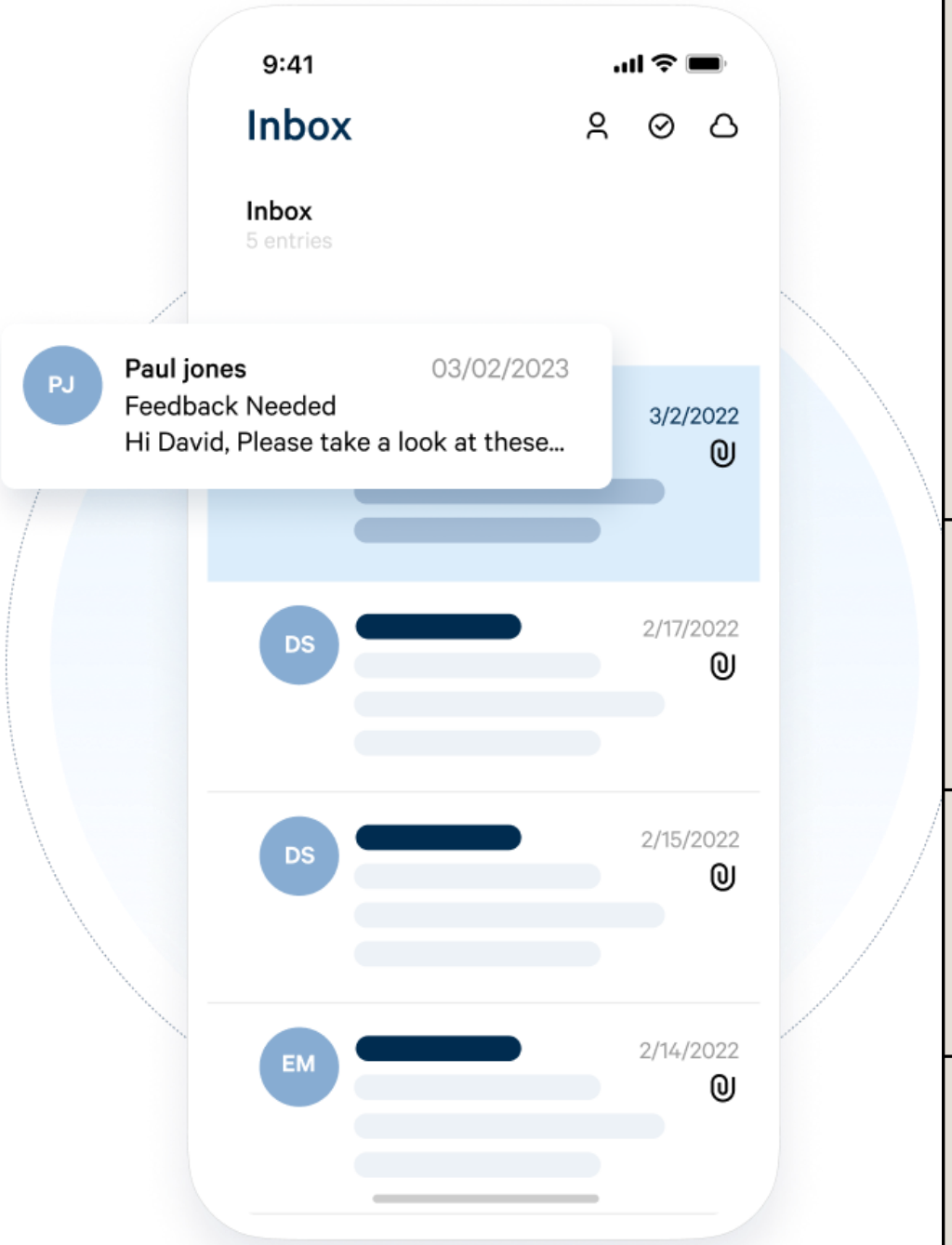

THANK YOU

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